Module 2
Skeletal System

This section discusses the skeletal system as it relates to directional references, identification of bones, spinal injury, and the foot/ankle complex.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Directional and landmark terminology
- Skeletal system
- Herniated disk
- Regions of the spine
- UESCA definition of *neutral spine*
- Spinal injuries
- Landmarks of pelvis
- Types of joints
- Joints of the foot/ankle and their functions

❖ DIRECTIONAL TERMINOLOGY

Within this certification program will be references to the planes of the body as well as directional terminology. There are three primary anatomical reference planes of the body:

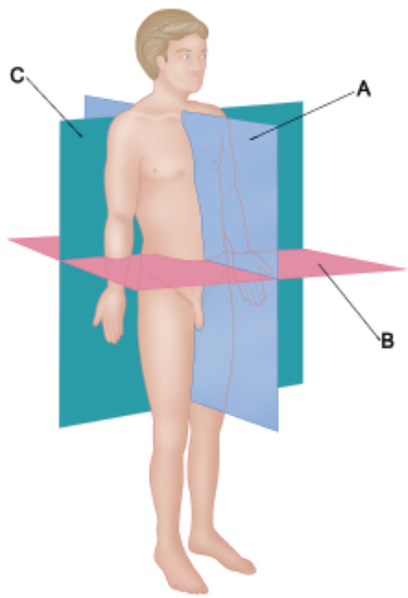


Figure 2.1 Planes of the Body

A. Sagittal: divides the body into right and left halves (e.g., forward motions such as running)

B. Transverse: divides the body into lower and top halves (e.g., rotating the upper body from left to right)

C. Frontal (Coronal): divides the body into front and back halves (e.g., side-to-side or lateral motions such as side steps)

When describing the relationship between body parts, movement of body parts, or the location of an external object to the body, the use of directional terms is necessary. Directional terms are paired, which represents opposing actions or locations. Some of the most common directional terms are:

Anterior: toward the front of the body (e.g., toes are anterior to the heel)

Posterior: toward the back of the body (e.g., the spine is posterior to the collarbone – or clavicle)

Medial: toward the midline of the body (e.g., the arm swing of a baseball player hitting a ball moves medially through the first half of the range of motion)

Lateral: away from the midline of the body (e.g., a backhand tennis return moves the arm laterally away from the body)

Superficial: toward the surface of the body (e.g., skin is superficial to subcutaneous fat)

Deep: inside the body, away from the surface (e.g., the intestines are deep relative to the skin)

Adduction (movement): toward the midline of the body (e.g., lowering arms to the legs from a starting position of the arms being extended out to the side)

Abduction (movement): away from the midline of the body (e.g., raising arms up to horizontal from a starting position by the side of the legs)

Flexion: movement that decreases the angle between body parts (e.g., bending forward at the waist flexes the abdominal muscles)

Extension: movement that increases the angle between body parts (e.g., going from a seated position to a standing position extends the knees)

Plantar Flexion: increase angle between lower leg and foot (e.g., women who wear high-heeled shoes are in a constant state of plantar flexion)

Dorsi Flexion: decrease angle between lower leg and foot (e.g., walking on the heels of the feet results in the feet being dorsiflexed)

Elevation: movement in the superior direction (e.g., when shrugging your shoulders, the shoulder blades – or scapulas – elevate)

Depression: movement in the inferior direction (e.g., when lowering your shoulders, the scapulas depress)

Medial Rotation (internal rotation): rotation toward the midline of the body (e.g., someone who is “knock-kneed” has medial rotation of the upper leg – or femur)

Lateral Rotation (external rotation): rotation away from the midline of the body (e.g., the leg is externally rotated if the feet are turned outward from the body's midline)

Unilateral: movement occurring on one side of the body (e.g., lifting just the right arm)

Bilateral: movement occurring on both sides of the body (e.g., lifting both arms)

Ipsilateral: on the same side as another structure (e.g., when lifting the right arm, it is ipsilateral to the right leg)

Contralateral: on the opposite side of another structure (e.g., when lifting the right arm, it is contralateral to the left leg)

Varus: outward angle of bone or joint (e.g., bowlegged)

Valgus: inward angle of bone or joint (e.g., knock–kneed)

- Varus and valgus often refer to substantial biomechanical abnormalities. However, these terms can also be used to describe normal biomechanical angles, such as the forefoot.

Protraction (anteriorly): moves forward (e.g., shoulders that are rounded to the front of the body are in a state of protraction)

Retraction (posteriorly): moves backward (e.g., squeezing the shoulder blades together results in retraction of the shoulder blades)

Proximal: point of attachment, closer to the center of the body (e.g., the femur is proximal to the tibia)

Distal: away from the point of attachment, farther from the center of the body (e.g., the tibia is distal to the femur)

Prone: facing downward (e.g., when people lie on their stomach, they are in the prone body position)

Supine: facing upward (e.g., when people lie on their back, they are in a supine position)

Inversion: turning inward (e.g., when the sole of the foot is facing medially, the foot is inverted)

Eversion: turning outward (e.g., when the sole of the foot is facing laterally, the foot is everted)

Rotation: circular motion around a fixed point (e.g., turning – or rotating – the torso in the transverse plane while in an upright, standing position)

Circumduction: the combination of abduction, adduction, flexion, and extension (e.g., moving a limb in a circular motion)

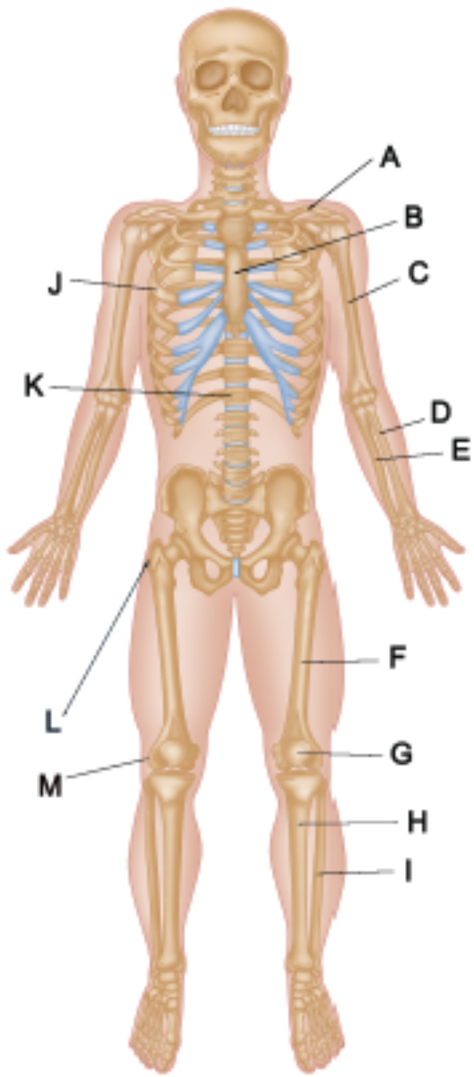
Superior: closer to the head (e.g., the torso is superior to the legs)

Inferior: farther away from the head (e.g., the feet are inferior to the knees)

ROM: Range **O**f **M**otion

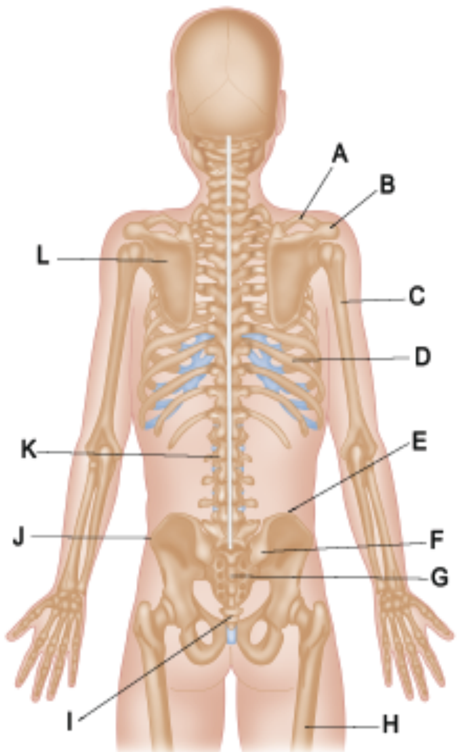
❖ SKELETAL SYSTEM

The adult human body is made up of 206 bones that constitute the skeletal system. At birth the body has more than 206 bones. However, some of these bones fuse together to create the adult skeletal system (212).



- **A.** Clavicle
- **B.** Sternum
- **C.** Humerus
- **D.** Radius
- **E.** Ulna
- **F.** Femur
- **G.** Patella
- **H.** Tibia
- **I.** Fibula
- **J.** Rib
- **K.** Vertebrae
- **L.** Greater Trochanter (top of femur)
- **M.** Lateral Femoral Epicondyle

Figure 2.2 Skeletal System – Anterior



A. Clavicle
B. Acromion Process
C. Humerus
D. Rib
E. Pelvic Crest
F. Posterior Superior Iliac Spine (PSIS)
G. Sacrum
H. Femur
I. Coccyx
J. Anterior Superior Iliac Spine (ASIS)
K. Vertebrae
L. Scapula

Figure 2.3 Skeletal System – Posterior

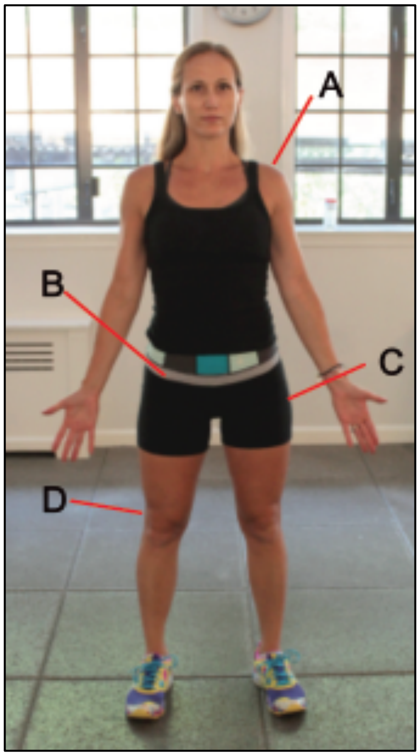
The skeletal system serves as the structural framework and levers of the body. The skeletal system can be broken down into two distinct regions: *axial* and *appendicular*.

Axial Skeleton:
Forms the central axis that is responsible for providing support to the appendicular skeleton. Comprises the skull, spine, ribs, and sternum (breastbone).

Appendicular Skeleton:
Represented by the limbs of the body and the bones that attach the limbs to the axial skeleton. Comprises the bones of the legs, arms, scapulas, clavicles, hands, feet, and pelvis.

LANDMARK TERMINOLOGY

This section defines skeletal points of the body that are often used for landmarks.



A: Acromion Process: Bony aspect of the shoulder blade (scapula) that extends above the shoulder joint. It is joined to the collarbone (clavicle) via the *acromioclavicular joint* (AC joint).

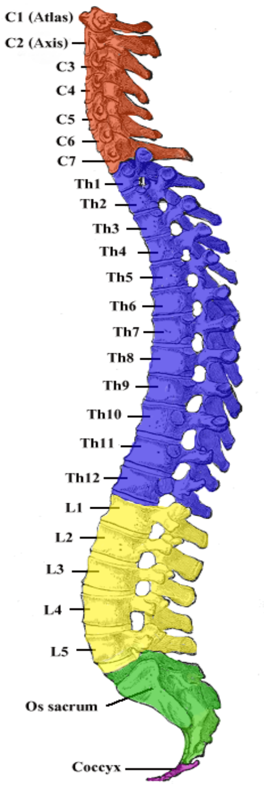
B: Anterior Superior Iliac Spine (ASIS): Noted later in the section regarding the pelvis, the ASIS is the anterior or forward most part of the pelvis. The ASIS is located by running the fingers down the side of the body until the top of the pelvic bone is felt. Then, continue to trace the pelvic bone inferiorly until the farthest anterior point of the ilium (pelvis) is felt.

C: Greater Trochanter: Lateral protrusion at the top of the femur

Figure 2.4 Body Landmarks

D: Lateral Femoral Epicondyle: A small, lateral, protruding aspect at the inferior aspect of the femur. Often used as a landmark, it is located to the lateral side of the knee. This landmark is often referenced as a source of pain from *iliotibial band syndrome* (IT band syndrome), as the epicondyle rubs against the IT band when running.

SPINE (i.e., Vertebral Column)



The spine is the critical element to the movement and stabilization of the body. The spine comprises 33 sections – or vertebrae – and between each vertebra is an *intervertebral disk* made of fibrocartilage. Intervertebral disks are what allow the spine to move and absorb compression. There are five distinct regions of the spine, and each spinal region assigns a number to individual vertebrae. The letter(s) in front of the number corresponds to the section of the spine (ex: T=thoracic). The numbers correspond to specific vertebrae, with the lowest number (i.e., 1) being the superior vertebrae in the section.

Cervical: (C1–C7) – Red
Thoracic: (T1–T12) – Blue
Lumbar: (L1–L5) – Yellow
Sacral: (S1–S5) – Green
Coccyx: (Co1–Co4) – Purple

Figure 2.5 Vertebral Column Sections

Image: Gray 111 – Vertebral column.png but colored. Copyright expired.

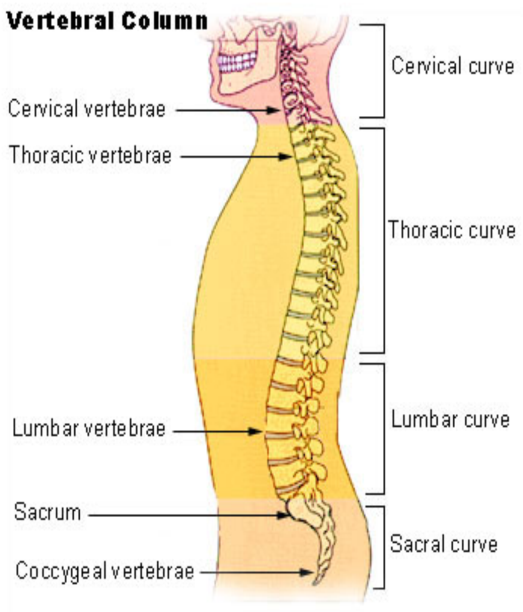
There are two primary types of curves in the spine:

1. **Kyphotic:** refers to an anterior curve of the spine
2. **Lordotic:** refers to a posterior curve of the spine

These terms are often used to denote extreme spinal curvatures. More specifically, these conditions are termed:

1. **Kyphosis:** extreme anterior curve in the spine (i.e., hunchback)
2. **Lordosis:** extreme posterior curve in the spine (i.e., sway back)

Below are the locations of the four spinal curvatures along with their respective curve type:



Cervical Curvature (Lordotic)
Thoracic Curvature (Kyphotic)
Lumbar Curvature (Lordotic)
Sacral Curvature (Kyphotic)

Figure 2.6 Vertebral Column Curvatures

http://training.seer.cancer.gov/module_anatomy/unit3_5_skeleton_divisions.html#

Neutral Spine Position

Neutral spine is a concept that has existed for a long time, and like other areas of exercise science, it has acquired multiple definitions. While many people refer to neutral spine as the perfect alignment of the spine, the reality is that neutral spine is different for everyone and is largely genetic and activity-based. For example, someone who slouches at a desk all day will have a very different neutral spine position than that of a professional runner.

For the purpose of this certification, the definition of neutral spine will be based on a neuromuscular approach:

> *Neutral spine is the position of the spine in which minimal neuromuscular activity is required to maintain a standing, relaxed posture.*

Using the preceding example, if you are working with a client whose neutral spine position is substantially slouched due to sitting at a desk all day, you will want to work with the person to create a more upright spinal position through ergonomics as well as strength and flexibility training.

The phrase “find your neutral spine” generally refers to the action taken to establish the ideal spinal position relative to the natural curves of the spine. This typically involves a predetermined alignment based on a lateral perspective from the head to the toes. Your client may or may not be able to establish or maintain this position.

Regarding neutral spine, the goal of a coach is to work with clients to help them establish and maintain a spinal position that improves posture, stability, and movement patterns, as well as maintains proper muscle length-tension relationships. As noted in the definition, the goal is for the client to be able to maintain these positions with minimal neuromuscular activity.

Lumbar Spine

Often referred to as the *low back*, this section of the spine is emphasized because of the important role it plays in stabilization and is the region of the spine where pain and injury are most often localized. While the cervical spine supports the weight of the head and the thoracic spine gains rigidity from the rib cage, the lumbar spine is largely dependent on muscular support to provide stabilization and mobility both superiorly and inferiorly. The lumbar spine is located between two stable structures, the ribs and the pelvis. Inefficient muscular stabilization of the lumbar spine is associated with lower back pain.

According to the *National Academy of Sports Medicine,* over $85 billion is spent annually to alleviate back pain (predominately lower back pain) in the US. Over 80 percent of all Americans will experience back pain within their lifetime (213).

Spinal Abnormalities

While it is natural to have spinal curvatures as described above, if the curves are excessive they are considered abnormal. Any number of factors such as osteoporosis, disease, accidents, obesity, pregnancy, or poor posture for extended periods can cause abnormal spinal curvature.

There are three classifications of spinal abnormalities:

1. **Lordosis:** excessive lumbar spinal curvature (hyperextension)
2. **Kyphosis:** excessive thoracic spinal curvature (hyperflexion)
3. **Scoliosis:** spinal curvature in the frontal plane (lateral curvature)

The two classifications of scoliosis are *structural* and *functional*. Functional scoliosis originates elsewhere in the body, such as a leg length discrepancy, whereas structural scoliosis originates because of a growth abnormality in the spine. Structural scoliosis often presents with rotation or twisting of the spine as well as lateral curvature. Visible signs of scoliosis are uneven hips and shoulders.

If you suspect your client has any of these three spinal abnormalities, you should refer the person to a physical therapist or physician for an evaluation.

Injury

A common injury of the spine is a **herniated disk**, also known as a ruptured disk, which occurs when an intervertebral disk is injured. Each disk is made up of two main components. The cartilage aspect of the disk is circular and is called the **annulus fibrosus**. The center of the disk is composed of a gelatinous substance called the **nucleus pulposus**. A herniated disk is caused when excessive stress placed on a disk causes the annulus fibrosis to crack and some of the nucleus pulposus leaks out (119). The leaked nucleus pulposus can put pressure on the spinal cord or spinal nerves. This pressure causes pain.

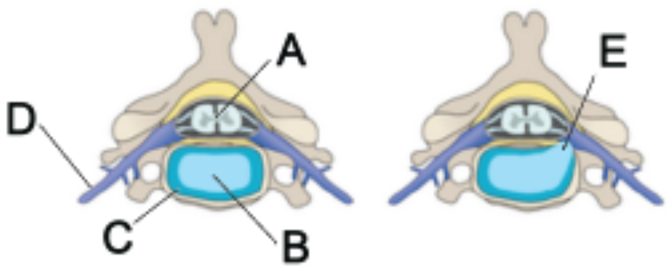
While surgery is occasionally recommended to repair a herniated disc, in most cases, surgery is reserved as a last course of action. Individuals with back pain caused by a herniated disc(s) typically see their pain reduced by conservative methods (i.e., pharmacology, rehabilitation, lifestyle changes) (784).

In a study that focused on individuals that had sciatica, which can be caused by disc herniation, after a 12-week period, 73% of the participants demonstrated ‘reasonable’ to ‘major’ improvement without surgery (785).

In respect to running, a recent study found that slow running (2 m/s) can actually increase the health of intervertebral discs via better disc composition and increased disc hypertrophy (786). Specifically, long distance runners (> 50km/week) had a greater positive effect than those that ran shorter distances (20-40 km/week).

In summation, surgery is typically only suggested for those individuals that do not respond positively to conservative methods.

In figure 2.7, the disk on the left is normal and the disk on the right exhibits a herniation as it shows the nucleus pulposus leaking out and putting pressure on the spinal nerve.



A. Spinal Canal
B. Nucleus Pulposus
C. Annulus Fibrosis
D. Spinal Nerve
E. Herniation of Disk

Figure 2.7 Herniated Disk

Annotated diagram of preconditions for anterior cervical discectomy and fusion.

A herniated disk is commonly referred to as a “slipped disk.” Technically, this is incorrect, as intervertebral disks cannot slip because they are attached to vertebrae (4).

Following are other causes of pain that involve the spine:

Spondylolisthesis
A slipped vertebra in relation to the one adjacent to it. Most commonly the vertebra moves anteriorly. However, it can move laterally and posteriorly (355).

Spinal Stenosis
Narrowing of the spinal canal because of degeneration, instability, or a lesion. The *spinal canal* is a vertical hole within the vertebrae that allows the spinal cord to pass through it (354).

Radiculopathy
Pain in the extremities (arms, hand, feet) that originates from compression of spinal nerves.

- **Sciatica:** A type of radicular pain. Presents with pain down the posterior and/or lateral aspect of the leg. This is more of a general term than a specific condition. Sciatica can manifest as a result of compression (i.e., herniated disk), spondylolisthesis, or muscle contraction.

If clients experience pain in either the back or extremities that is not simply resolved by a change in body position or rest, they must be referred to a physician or physical therapist.

PELVIS

The pelvis is a very important structure in the human body. The pelvis is commonly referred to as the hips and separates the lower body from the torso. Because of its location, movement of the pelvis affects both the upper and lower body. From a skeletal perspective, the pelvis represents the base of the spine as well as the location where the femurs insert.

Mechanically speaking, the primary purpose of the pelvis is to support the upper body and to transfer the energy from the legs to the spine and upper body. The pelvis protects reproductive and digestive organs of the lower torso.

Key Areas of the Pelvis

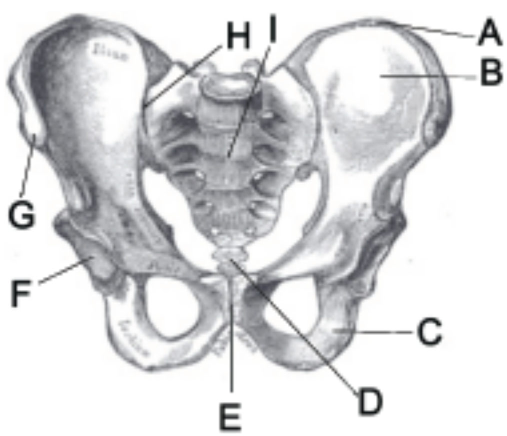


Figure 2.8 Pelvic Landmarks

20th U.S. edition of Gray's Anatomy of the Human Body (copyright expired)

A. **Iliac Crest**
   The most superior aspect of the ilium is called the *iliac crest* which is commonly used as an anatomical landmark.

B. **Ilium**
   Located on either side of the sacrum, the ilium is the largest bone(s) in the pelvis. The primary purpose of the ilium is to protect internal organs.

C. Ishium

Often referred to as the *sits bones*, the ishium bones are located where one's body weight is distributed in a seated position.

D. Coccyx

Commonly referred to as the *tailbone*, the coccyx is the lowest part of the vertebral column.

E. Pubic Symphysis

Connects the two sides of the lower pelvis. This joint is a semimovable joint and is connected by ligaments and cartilage. The pubic symphysis expands as the distance between the legs expands. Pain in the pubic symphysis can occur when the joint does not return to the normal position (i.e., is misaligned).

F. Acetabulum

The location of the pelvis that the head of the femur sits in.

G. Anterior Superior Iliac Spine (ASIS)

The ASIS is the forward most part of the pelvis. The ASIS is located by running the fingers down the side of the body until the top of the pelvic bone is felt. Then, continue to trace the pelvic bone inferiorly until the farthest anterior point of the ilium is felt.

H. Sacroiliac Joint

Often termed the *SI joint.* The SI joint is the connection between the sacrum and the ilium, located on both sides of the spine. The SI joint is stabilized by strong ligaments. The purpose of the joint is to absorb shock. The joint "locks" during walking and running to provide a solid base of support during the foot push-off aspect of the gait cycle (245). Inflammation of the SI joint can be caused by too much or too little motion and typically presents with pain in the SI joint and low back region (244).

I. Sacrum

Comprises the base of the spine and is made up of five fused vertebrae. Located between the lowest lumbar vertebrae (L5) and the coccyx.

JOINTS

There are three primary classifications of joints within the body:

1. **Fibrous** (e.g., joints of the skull)
2. **Cartilaginous** (e.g., intervertebral disks)
3. **Synovial** (e.g., shoulder joint)

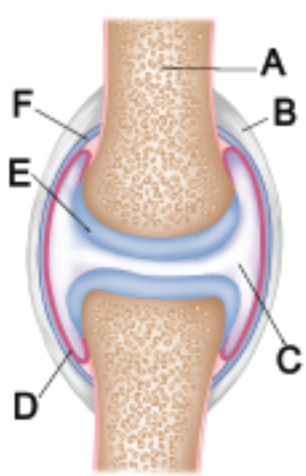
Joints can be further classified by the amount of movement. The three functional classifications of joint movement are:

1. **Immoveable:** fibrous joints
2. **Slightly Moveable:** cartilaginous joints
3. **Freely Moveable:** synovial joints

Of these three classifications of joints, synovial joints are by far the most prevalent in the body.

Synovial joints are made up of the following components:

1. **Synovial fluid**: a collagen-like substance that surrounds a joint
2. **Synovial membrane** or **synovial cavity**: the inner layer of a joint capsule that secretes synovial fluid – a lubricating liquid
3. **Cartilage**: pads the ends of the bones
4. **Joint capsule**: fibrous sac that contains the synovial membrane/fluid



A. Bone
B. Ligament
C. Synovial Fluid
D. Synovial Membrane
E. Cartilage
F. Joint Capsule

Figure 2.9 Joint Landmarks

‘Cracking’ Joints

Most commonly associated with the knuckles, joints can make a cracking or popping noise when moved into particular positions. This issue is noted because clients often ask if cracking joints mean they are injured or if there is something wrong with them. This noise does not signify that an injury has occurred. As noted previously, a synovial joint consists of a fluid-filled cavity. When the bones of a joint are pulled away from each other (e.g., knuckles cracking), the synovial cavity space increases in size.

OLD THEORY

It was previously thought that the cracking was caused by a decrease in synovial fluid pressure within the cavity. This decrease in fluid pressure was thought to cause bubbles to form within the cavity through a process called *cavitation*. Therefore if the bones of the joint continued to be pulled away from each other, the pressure in the cavity would drop low enough that the bubbles pop, thus causing the audible cracking noise (246).

NEW THEORY

Research by Kawchuk et al. found that the cracking noise emitted by joints was not due to a bubble popping within the joint but rather the rapid creation of a temporary cavity within the joint (termed *tribonucleation*), thus debunking the prior theory (770). This new theory was discovered by viewing a real-time Cine MRI of the joint separating and cracking. Regardless, of the actual cause of the cracking, there seems to be no danger in cracking joints.

ANKLE/FOOT COMPLEX

The foot and ankle complex is a very busy area. This complex has 26 bones, 33 joints, and over 100 muscles, tendons, and ligaments.

Skeletal Structure

- **Leg Bones:** tibia, fibula
- **Ankle Bone:** talus
- **Heel Bone:** calcaneus

- **Midfoot Bones (foot arch):** cuboid, navicular, cuneiform (3)
- **Forefoot Bones:** metatarsals, phalanges (toes)

Figure 2.10 Foot/Ankle Bones (side of foot)

The foot and ankle complex plays a major role in running. In spite of this, it is often an overlooked and misunderstood part of the body, specifically in regard to its mechanics.

While not noted in figure 2.10, there are typically two small, pea-sized bones (sesamoid) located underneath the first metatarsal bone (behind the big toe). These bones are embedded into a tendon and functionally are used to provide leverage when walking and running. Because of the stress placed on these bones when running, they can break and/or the surrounding tendon can become irritated, called **sesamoiditis**. Sesamoiditis is discussed in greater detail in the *Injury and Illness* module.

Joints

Two primary joints function within the ankle.

1. **Talocrural Joint:** Connects the tibia and fibula with the talus bone in the foot. This is a hinge joint and provides *dorsi flexion* and *plantar flexion*.

2. **Subtalar Joint:** Connects the *talus bone* and *calcaneous bone*. This is a gliding joint and allows inversion and eversion of the ankle to occur. The *Achilles tendon* covers the subtalar joint when viewing the leg from a posterior aspect.

Figure 2.11 Foot/Ankle Joints (top of foot)

Two primary joints in the foot affect gait and foot stability.

Midtarsal (Transverse Tarsal and Talocalcaneonavicular)

This joint spans the arch of the foot and primarily controls inversion and eversion of the foot (figure 2.12). This joint ranges from flexible to rigid based on what aspect of the support and drive phase the foot is in. As the foot moves under the body's center of gravity and prepares to push off, the joint locks up and becomes rigid to provide a stable base of support to push off from (214).

Tarsometatarsal

Like the midtarsal joint, this joint also spans across the width of the foot arch. However, it is located posterior to the metatarsals, effectively at the junction between the midfoot and forefoot. This joint has more stability and less range of mobility than the midtarsal joint. Figure 2.12 illustrates the bottom of the foot in respect to the midtarsal and tarsometatarsal joints.

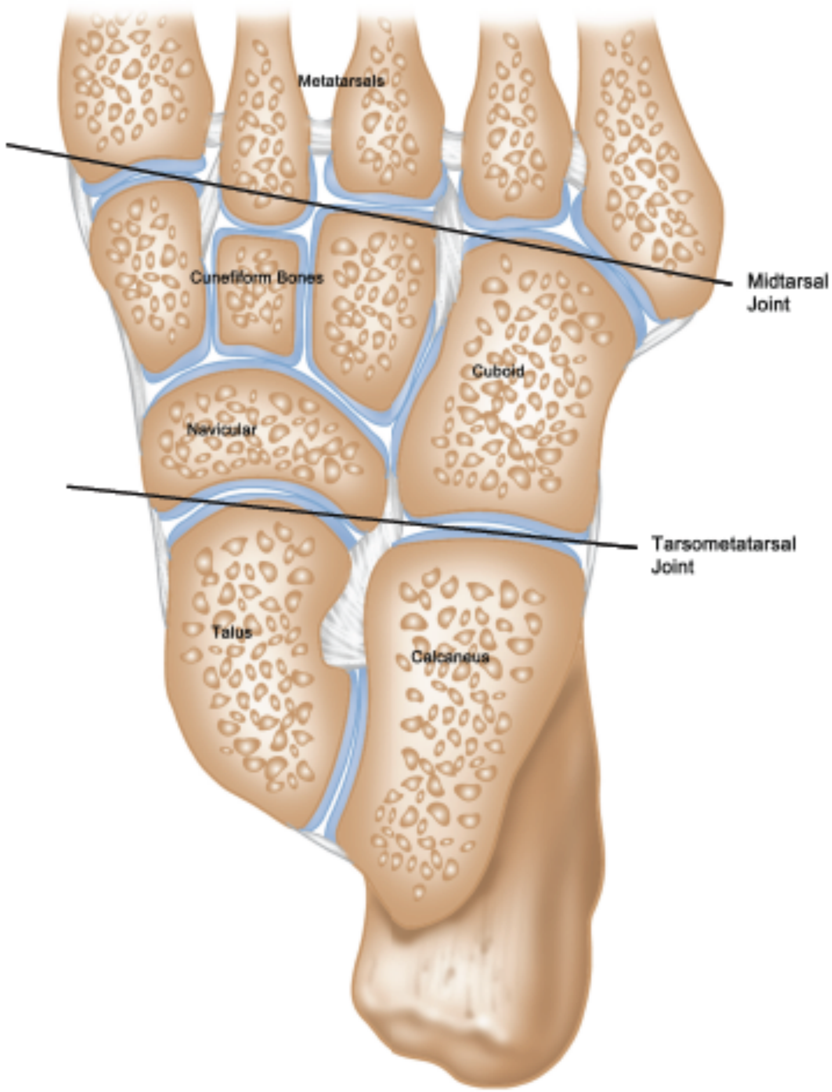


Figure 2.12 Foot Joints (bottom of foot)

The foot and ankle joints act together to allow and restrict motion so that proper mechanics can occur.

GENETIC FACTORS

It is common knowledge that weight-bearing activities such as running and weight lifting are beneficial for increasing bone density. The more stress put on a bone(s), the stronger it will become. However, bone strength has also been found to have a genetic component. A 2010 study by Libby Cowgill found that differences in bone strength are evident as early as one year of age (491). So while bone can become denser through training, there is also a large genetic factor.

In the book *The Sports Gene*, the author cites research by Francis Holway that compares the human skeleton to that of a bookcase (466). The stronger and heavier a bookcase is, the more books it can carry. However, Holway's research found that there is a limit to how much muscle weight a skeleton can carry. Specifically, 2.2 pounds of bone can support a maximum 11 pounds of muscle, or a five-to-one ratio of muscle to bone. Once this ratio is reached, any additional weight gain is typically fat versus muscle (466).

❖ SUMMARY

- The body is divided into three planes: *sagittal, frontal,* and *transverse.*
- The spine is made up of five sections: *cervical, thoracic, lumbar, sacral,* and *coccyx.*
- The skeletal system can be broken down into two distinct regions: *axial* and *appendicular*.
- The spine is made up of 33 vertebrae.
- "Cracking joints" is not indicative of injury.
- Kyphosis refers to an anterior curve of the spine.
- Lordosis refers to a posterior curve of the spine.
- *Neutral spine* is the position of the spine in which minimal neuromuscular activity is required to maintain a standing, relaxed posture.
- The lower back (i.e., lumbar spine) is the area of the spine most commonly associated with pain.
- Synovial joints are the most common type of joint in the body.
- A herniated disk causes the nucleus pulposus to leak out and results in pressure on the spinal nerve.
- Sciatica typically causes pain to radiate down the leg.

- The primary purpose of the pelvis is to support the upper body and to transfer the energy from the legs to the spine and upper body.
- Weight-bearing activities increase bone density.
- There is a genetic component to bone strength.
- The three primary classifications of joints are:
    - Fibrous
    - Cartilaginous
    - Synovial (most prevalent joint type)
- Four major joints in the foot/ankle affect gait and foot stability:
    - Talocrural (ankle)
    - Subtalar (ankle)
    - Midtarsal (foot)
    - Tarsometatarsal (foot)